

Decision summaries



RADLogics, Inc.
% John J. Smith, M.D., J.D.
Partner
Hogan Lovells US LLP
555 13th Street NW
WASHINGTON DC, 20004

April 8, 2020

Re: K191300
Trade/Device Name: AIMI-Triage CXR PTX
Regulation Number: 21 CFR 892.2080
Regulation Name: Radiological computer aided triage and notification software
Regulatory Class: Class II
Product Code: QFM
Dated: March 9, 2020
Received: March 9, 2020

Dear Dr. Smith:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cdrtxt/cfpm/pma.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807), labeling (21 CFR Part 801), medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for

U.S. Food & Drug Administration
2000 New Hampshire Avenue
Silver Spring, MD 20910
(800) 368-0711

Device Performance by Dataset and Region

Category	N	Sensitivity	Specificity	AUROC
NIH (US)	147	97.6% (93.2,99.2)	90.6% (84.5,94.7)	0.987 (0.973,0.999)
PADCHEST (OUS)	153	85.3% (79.0,90.8)	89.7% (83.6,93.9)	0.949 (0.918,0.979)

Device Performance by Scanner Spatial Resolution

Spatial Resolution Category	N	Sensitivity	Specificity	AUROC
High range				
<0.145 (above 3.45 lp/mm)	133	89.5% (83.0,94.1)	85.1% (77.7,90.6)	0.976 (0.944,0.999)
Mid range				
0.145-0.170 (2.95-3.45 lp/mm)	88	92.6% (84.3,96.7)	93.4% (85.7,97.4)	0.983 (0.961,0.999)
Low range				
>0.170 (below 2.95 lp/mm)	64	93.1% (84.8,98.3)	91.4% (80.7,96.5)	0.946 (0.911,0.980)

Thus, the device also performs comparably to the predicate in terms of time savings. The AIMI-Triage CXR PTX has a safety and effectiveness profile that is similar to the predicate device.

Conclusions

The AIMI-Triage CXR PTX has the same intended use and very similar indications, technological characteristics, and principles of operation as its predicate device. The minor differences in indications do not alter the device's intended use and do not affect its safety and effectiveness when used as labeled. In addition, the minor technological differences between the device and its predicate raise no new issues of safety or effectiveness. Performance data further demonstrate that the AIMI-Triage CXR PTX is as safe and effective as the HealthPNX (K190362). Thus, the subject device is substantially equivalent to the identified predicate.

Substantial Equivalence Table

	AIMI-Triage CXR PTX	HealthPNX (K190362)
Intended Use / Indications for Use	The AIMI-Triage CXR PTX Application is a notification-only triage workflow tool for use by hospital networks and clinics to identify and help prioritize chest X-rays acquired in the acute setting for review by hospital radiologists. The device operates in parallel to and independent of standard of care image interpretation workflow. Specifically, the device uses an artificial intelligence algorithm to analyze images for features suggestive of moderate to large sized pneumothorax; it makes case-level	The Zebra Pneumothorax device is a software workflow tool designed to aid the clinical assessment of adult Chest X-Ray cases with features suggestive of Pneumothorax in the medical care environment. HealthPNX analyzes cases using an artificial intelligence algorithm to identify suspected

	AIMI-Triage CXR PTX	HealthPNX (K190362)
	output available to a PACS/workstation for worklist prioritization or triage. Identification of suspected cases of moderate to large sized pneumothorax is not for diagnostic use beyond notification.	Findings. It makes case-level output available to a PACS/workstation for worklist prioritization or triage. HealthPNX is not intended to direct attention to specific portions or anomalies of an image. Its results are not intended to be used on a stand-alone basis for clinical decision-making nor is it intended to rule out Pneumothorax or otherwise preclude clinical assessment of X-Ray cases.
	The AIMI-Triage CXR PTX Application is limited to analysis of imaging data as a guide to possible urgency of adult chest X-ray image review, and should not be used in lieu of full patient evaluation or relied upon to make or confirm diagnoses. Notified radiologists are responsible for engaging in appropriate patient evaluation as per local hospital procedure before making care-related decisions or requests. The device does not replace review and diagnosis of the X-rays by radiologists. The device is not intended to be used with plain film X-rays.	
Classification	21 C.F.R. § 892.2080, Product Code QFM	
User	Radiologist	
Anatomical Region	Lung	
Clinical Condition	Pneumothorax	
Modality	Chest X-ray	
Segmentation of region of interest	No; device does not mark, highlight, or direct users' attention to a specific location in the original image	
Alteration of original image	No	
Relation to standard of care workflow	Independent/parallel; no cases are removed from worklist queue	
Algorithm	Artificial intelligence algorithm with database of images	
Notification / Prioritization	Yes	
Alert to Finding	Passive notification – flagged for review	
Where Results are Received	PACS / Workstation	

Regulatory policies and guidances

- 21 Code of Federal Regulations (CFR)
- FDA Coding Resources for Medical Device Reports
- FDA guidances on “Good Machine Learning Practice” and “Recommendations for the Use of Clinical Data”
- ...

Medical Device Reports from MAUDE

GE HEALTHCARE AUSTRIA GMBH & CO OG; DIAGNOSTIC ULTRASOUND TRANSDUCER

Model Number IC9-RS

Device Problem Poor Quality Image (1408)

Patient Problem No Clinical Signs, Symptoms or Conditions (4582)

Event Type malfunction

Event Description

An inspection test was performed as part of correction and removal initiated by ge healthcare on 29-dec-2023 (us-fda recall no.Z-0865-2024), and it was concluded the ic9-rs diagnostic ultrasound probe was identified as having a component malfunction described in us-fda recall no.Z-0865-2024. There was no patient involvement.

Manufacturer Narrative

Legal manufacturer: hcs kretz - tiefenbach 15 austria zipf oberosterreich, 4871.Ge healthcare reported a field modification for this ic9-rs double image malfunction per 21 cfr 806 on 29-dec-2023. The us-fda recall number is z-0865-2024.Customers were sent a letter explaining the issue and requesting the customer to perform an inspection test to determine if the probe is malfunctioning.Ge healthcare has determined the cause of the malfunctioning probe to be a probe component.To correct this issue, ge healthcare has replaced this malfunctioning probe.

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